CIS COLLEGE
IRAQ INTERNATIONAL

AI DIPLOMA / FINAL DATA ANALYTICS PROJECT

Industrial Sensor Data Quality Analysis

Factory Sensor Telemetry
Data Quality · Cleaning · Exploratory Analysis

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Field : Data Analytics & Data Preprocessing

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Tools & Technologies : Python · Pandas · NumPy · Matplotlib · Jupyter Notebook

8,000	12	3	4
Records after cleaning	Sensor devices	Operational sites	Analytical charts

Scope of the study : Assess the quality of synthetic sensor data, apply verifiable cleaning procedures, and examine patterns and trends without asserting confirmed engineering faults.

REPORT STRUCTURE / CONTENTS

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> The indicators are descriptive findings from synthetic data. They should not be treated as independent evidence of actual equipment failure or as approved engineering safety limits.

1. Executive Summary

This project assesses the quality of synthetic industrial sensor data, cleans formatting inconsistencies and duplicate records, and then examines the measurements to identify patterns and differences across devices and sites over the observation period. The analysis uses Python, Pandas, NumPy, and Matplotlib. Its processing stages and validation tests are documented in two independent Jupyter notebooks.

The original dataset contains **8,000 records and 8 columns**, representing measurements from **12 devices at 3 sites**. For data-processing practice, a working raw dataset was prepared with deliberately introduced data-quality issues, increasing its size to **8,020 records**. Cleaning produced **8,000 records**, while retaining missing values and statistical outliers when there was insufficient evidence to justify imputation or removal.

The assessment identified **20 fully duplicated records** and **40 missing measurement values**, along with formatting inconsistencies in selected columns. The analysis found a positive descriptive association between temperature and vibration, a concentration of most vibration outliers in two devices, and **8 long temporal gaps** also present in the original dataset. Additional testing showed that the number of observations flagged as outliers depends on how statistical limits are defined.

Principal conclusion: The project completed its objectives in data-quality assessment, cleaning, validation, and exploratory analysis, but its findings remain specific to synthetic data. Statistical indicators do not establish actual mechanical failures or predict future equipment failures.

2. Dataset Description and Study Scope

2.1 Data Source and Characteristics

The study uses Datanemics' Factory Sensor Telemetry dataset, a **synthetic dataset** that simulates industrial sensor measurements. The source identifies its license as CC0 1.0.

The data cover **1 January through 28 February 2023**. The 12 devices are distributed across three sites:

- `NOR` — four devices
- `RIV` — four devices
- `SLT` — four devices

The original synthetic dataset already contains unusual patterns and temporal gaps. This report therefore distinguishes **issues deliberately introduced into the training copy** from **characteristics already present in the original synthetic data**.

2.2 Dataset Files

File	Purpose	Records
`original_data.csv`	Unmodified reference copy of the original synthetic dataset	8,000
`raw_data.csv`	Working raw copy with deliberately added data-quality issues	8,020
`cleaned_data.csv`	Dataset resulting from cleaning and validation	8,000

The project does not include a standalone script that regenerates the introduced quality issues from the original dataset. Both notebooks can be rerun from the provided `raw_data.csv`; `original_data.csv` remains an independent comparison reference.

2.3 Dataset Columns and Measurement Units

Column	Meaning	Unit or value type
`reading_at`	Measurement timestamp	UTC according to the source description
`device_id`	Device identifier	Text identifier
`site`	Device site	`NOR` / `RIV` / `SLT`
`status`	Recorded status category	`OK` / `WARN` / `FAULT`
`temp_c`	Temperature	°C
`humidity_pct`	Relative humidity	%
`vibration_mm_s`	RMS vibration velocity	mm/s
`rpm`	Rotational speed	Revolutions per minute

Technical note: The source describes timestamps as UTC, but individual CSV timestamps do not contain explicit timezone information. The project retains those values as supplied, without converting them to local time. RMS means **Root Mean Square** in the description of vibration velocity.

2.4 Research Questions

The study is guided by a primary question and related sub-questions covering data-quality assessment, cleaning, and the analysis of operational measurements.

Primary research question:

How reliable is the factory sensor dataset, and what patterns and trends can be observed in measurements across devices, sites, and the study period?

Supporting research questions:

1. Which data-quality issues are present in the working raw dataset, and how extensive is each issue?
2. Which cleaning actions can be performed safely, and how can their results be validated?
3. How do temperature, vibration, humidity, and rotational speed differ across devices and sites?
4. What temporal trends, gaps, and statistical outliers are evident in the data?
5. What are the limits of interpreting these patterns, and what additional requirements would be needed to use the data in machine learning later?

3. Methodology

The project follows an integrated workflow: understanding the raw dataset, assessing its quality, cleaning issues that can be addressed reliably, validating the results, and carrying out exploratory analysis and visualization. Two notebooks separate these responsibilities and allow the processing decisions and findings to be reviewed independently.

3.1 Data Quality Assessment

The audit examines dataset dimensions, column types, fully duplicated records, device-and-time keys, missing values, inconsistent category formatting, and rotational-speed formatting. Descriptive statistics and the **interquartile range (IQR)** are also used to identify statistically unusual measurements.

3.2 Data Cleaning

Cleaning is implemented in `02_data_cleaning_validation.ipynb`, beginning with the working raw data and leaving the original reference file unchanged. Operations include removing exact duplicates, parsing timestamps, standardizing text categories, correcting numerical formatting, sorting records, and saving the cleaned dataset.

3.3 Data Validation

The notebook uses `assert` checks to validate record counts, the uniqueness of device-and-time keys, column types, category consistency, and the locations of missing values. The saved dataset is also reloaded and checked for content equivalence and expected data types under the specified reading settings.

An additional check compares the analytical dataset used in the quality-analysis notebook with `cleaned_data.csv`, after standardizing formatting and sorting the records. This test passed for **all 8,000 records and every column**, using a small numerical tolerance and allowing differences in the in-memory representation of data types.

3.4 Exploratory Data Analysis

The analysis is implemented in `01_data_quality_analysis.ipynb`. It includes descriptive statistics; filtering and sorting; aggregations with `groupby()`; pivot tables; and the examination of correlations, daily trends, and temporal gaps.

The derived field `reading_day` is created from the timestamp for daily aggregation within the analytical workflow. It is not added to the raw or cleaned CSV file.

3.5 Execution Environment and Reproducibility

Both notebooks were successfully rerun from fresh Jupyter kernels using **Python 3.14.7** and the versions documented in `requirements.txt`: **Pandas 3.1.0**, **NumPy 2.5.4**, **Matplotlib 3.11.2**, and **ipykernel 7.4.0**.

The notebooks use relative paths from the `notebooks` directory. Rerunning the cleaning notebook rewrites `cleaned_data.csv`, an effect that should be considered when assembling the submission package. The detailed execution procedure is described in `README.md`.

4. Data Quality Findings

4.1 Issues Identified in the Working Raw Dataset

Assessment item	Result
Raw dataset size	8,020 rows × 8 columns
Fully duplicated rows	20
Missing measurement values	40
Missing values per measurement column	10
Site-code formatting inconsistencies	12
Status-code formatting inconsistencies	12
RPM values containing thousands separators	12
Invalid timestamps	0
Long temporal gaps identified	8

Interpretation: Exact duplicates, thousands separators, and category-formatting inconsistencies can be corrected directly. Missing values, temporal gaps, and statistical outliers, however, should not be replaced or discarded without reliable supporting information.

4.2 Distribution of Missing Values

The dataset contains **40 missing values**, distributed equally across four numeric measurement columns:

Column	Missing values
`temp_c`	10
`humidity_pct`	10
`vibration_mm_s`	10
`rpm`	10
Total	**40**

By site, **26** missing values occur at `NOR`, **8** at `RIV`, and **6** at `SLT`. This distribution is uneven, but it does not establish any specific operational cause.

5. Data Cleaning and Validation

5.1 Cleaning Decision Log

Issue	Action taken	Rationale
20 exact duplicate records	Remove duplicates	Prevent counting a reading more than once
Timestamps stored as text	Parse `reading_at` as datetime	Enable chronological sorting and calculations
Inconsistent `site` and `status` formats	Trim whitespace and standardize case	Ensure consistent grouping and categories
Thousands separators in `rpm`	Remove separators, then parse numerically	Support correct numeric calculations
Missing values	Retain without imputation	Avoid inventing undocumented measurements
Statistical outliers	Retain without automatic deletion	No engineering evidence that they are errors
Temporal gaps	Do not generate replacement records	No independent record of unobserved measurements

After processing, records are sorted by timestamp and device identifier and saved to `cleaned_data.csv`.

5.2 Cleaning Results

Validation indicator	Result after cleaning
Number of records	8,000
Number of columns	8
Fully duplicated records	0
Missing values	40, retained without imputation
Site and status formatting	Standardized
Numeric and datetime types after reload	Passed validation
Saved file versus cleaning output	Passed equivalence test
Analytical versus cleaned data after normalization	Passed equivalence test

Methodological clarification: CSV does not independently preserve programming-language data types; those types are determined when the file is reloaded with specified settings. The equivalence tests also allow limited differences in numeric precision and type representation. Passing these tests does not establish the engineering accuracy of the sensor measurements.

6. Exploratory Analysis Results

6.1 Filtering and Aggregation

The analysis applies filters with distinct objectives, including isolating `FAULT` readings, temperature readings above the 90th percentile, and readings from `NOR`. Aggregations and comparisons are conducted by site, device, and status, alongside device rankings and a pivot table of mean vibration.

Filtering indicator	Result
Records with `FAULT` status	124
90th percentile of temperature	45.151 °C
Temperature readings above the 90th percentile	799
Records at site `NOR`	2,823

Of the **799 temperature readings above the 90th percentile**, **414** have status `OK`, **274** have status `WARN`, and **111** have status `FAULT`. Thus, comparatively high temperature alone does not map directly to the recorded operating-status categories.

6.2 Statistical Outliers — IQR

The original outlier assessment uses a single set of IQR limits for each measurement variable, calculated from all valid observations of that variable after removing duplicate records.

Measurement	Outlier readings	Approximate share of 8,000 records
Temperature	89	1.1%
Humidity	49	0.6%
Vibration	677	8.5%
Rotational speed	61	0.8%

Of the **677 vibration outliers**, **650** occur in two devices: **419 at** `MCH-04` and **231 at** `MCH-10`. Together they account for approximately **96%** of vibration outliers under this definition.

This concentration identifies a pattern worth investigating; it does not imply that every flagged reading is incorrect or that either device experienced a confirmed mechanical failure.

6.3 Comparison of Two Temperature Outlier Methods

An additional comparison examines how the **reference population used to derive IQR limits** affects outlier classification. Both new approaches derive their limits from records with status `OK`, then apply those limits to all temperature readings:

- **Method 1:** One global set of limits derived from `OK` readings across all devices.
- **Method 2:** Separate limits for each device, derived from that device's `OK` readings.

Comparison outcome	Readings
Outliers under global `OK`-based limits	133
Outliers under device-specific `OK`-based limits	89
Outliers under both methods	88
Outliers under global limits only	45
Outliers under device-specific limits only	1
Records classified differently	46

Interpretation: The difference in total outlier counts is **44 readings**, while **46 individual records** receive different classifications because one reading is flagged only by the device-specific approach. Device `MCH-06` has two differently classified readings even though the two methods identify the same total number of outliers for that device.

Important distinction: The **89** temperature outliers in the original IQR analysis are based on all valid temperature readings. The **89** in Method 2 above result from device-specific limits calculated using `OK` readings. Equal totals do not imply identical methods or identical flagged records.

6.4 Mean Vibration Across Sites

Site	Mean vibration
`NOR`	3.3592 mm/s
`RIV`	2.8044 mm/s
`SLT`	2.6043 mm/s

`NOR` has the highest overall mean, but its mean falls to **2.7033 mm/s** when readings from `MCH-04` and `MCH-10` are excluded **for analytical comparison only**. The resulting mean is lower than the mean for `RIV`.

This illustrates how a site-level mean can be strongly influenced by particular devices. It would be inappropriate to assume that every device at `NOR` exhibits the same vibration behavior. No readings from these two devices were removed from the original or cleaned data files.

6.5 Temperature–Vibration Correlation

Reading subset	Pearson correlation coefficient
Full analytical dataset	0.6232
Records with `OK` status only	0.3093

The positive coefficients indicate that temperature and vibration tend to rise together in the studied data. However, the lower coefficient within `OK` readings shows that the strength of this association depends on the subset being analyzed.

Correlation does not demonstrate that an increase in temperature causes an increase in vibration, or vice versa. It is also insufficient on its own to diagnose equipment condition.

6.6 Consistency of Status Categories with Vibration

An exploratory test compares the recorded `status` categories against two vibration thresholds: **4.2** and **5.5 mm/s**. All **7,990 records** with non-missing vibration values match the tested classification rule, with no mismatches.

Status	Records matching the tested rule
`OK`	7,387
`WARN`	481
`FAULT`	122
Total testable records	**7,990**

The remaining **10 records** cannot be checked against this rule because their vibration values are missing, even though they have recorded statuses: **7** `OK`, **1** `WARN`, and **2** `FAULT`.

Limit of interpretation: The test demonstrates consistency with the investigated rule for records having valid vibration values. It does not prove that this was the original rule used to generate the labels or that the thresholds constitute an accepted engineering standard for faults.

6.7 Temporal Gaps and Device Coverage

The analysis identifies **8 long temporal gaps** affecting four devices at site `RIV`:

`MCH-02` · `MCH-05` · `MCH-08` · `MCH-11`

Each device has two gaps. Four gaps last **4 days and 22 hours**, while the other four last **5 days**. They occur during two periods in January and February 2023 and are confirmed to exist in `original_data.csv`, before any training-specific quality issues were introduced.

The dataset covers **59 calendar days**. Readings occur on all 59 days for eight devices, while each of the four affected devices has readings on **51 days**. Daily coverage alone does not guarantee that every expected reading within a day is available.

Assuming a typical interval of **two hours between consecutive readings from the same device**, the estimated number of potentially unrecorded observations within those gaps is **468**. This is conditional on the observed sampling pattern, not a verified count of missing records or evidence of why recording was interrupted.

6.8 Daily Vibration Trends

The daily mean measurements indicate an upward trend for the following devices:

Device	Approximate daily slope
`MCH-04`	0.0540 mm/s/day
`MCH-10`	0.0487 mm/s/day

A threshold of **0.02 mm/s/day** was selected as an exploratory screening criterion for clearer upward trends. Only these two devices exceed it in the reported test.

Methodological note: This threshold is an analytical choice, not an approved engineering limit. An upward slope does not, by itself, establish mechanical deterioration or imminent failure, especially in a synthetic dataset.

7. Data Visualization

Four charts were created from the analysis and saved in the `charts` directory. They provide visual support for the numerical findings but do not replace the validation tests or methodological evidence.

7.1 Temperature Distribution

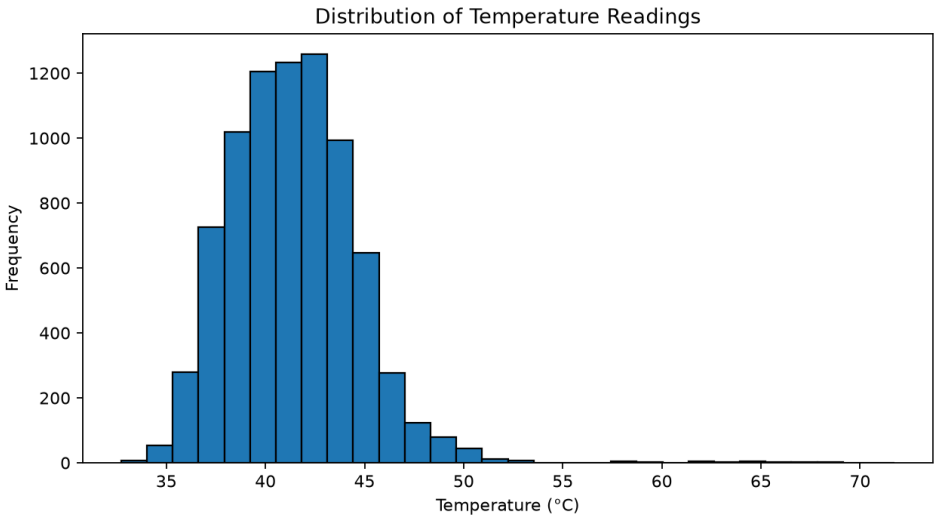


Figure 1: Shows the temperature distribution and areas of greatest concentration, helping identify the overall spread and values distant from the densest part of the distribution.

7.2 Mean Vibration by Site

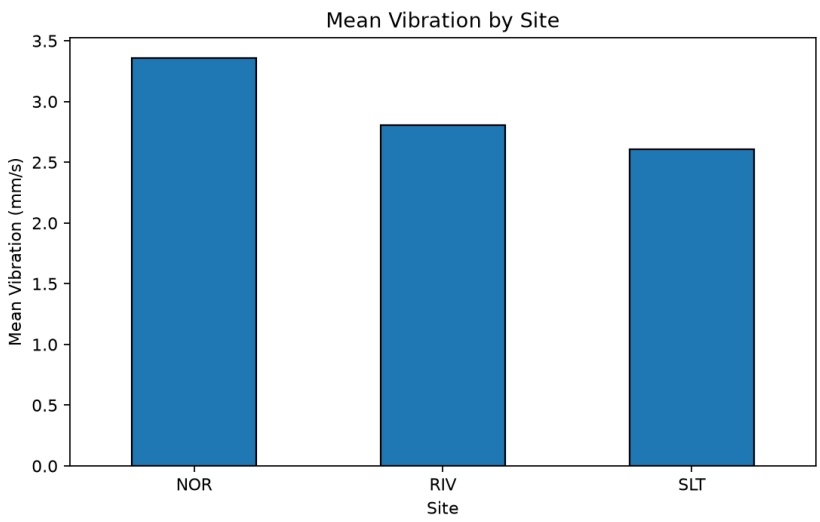


Figure 2: Illustrates differences in the overall site means and complements the analysis showing how the two devices with higher readings influence the mean for `NOR`.

7.3 Daily Vibration Trend

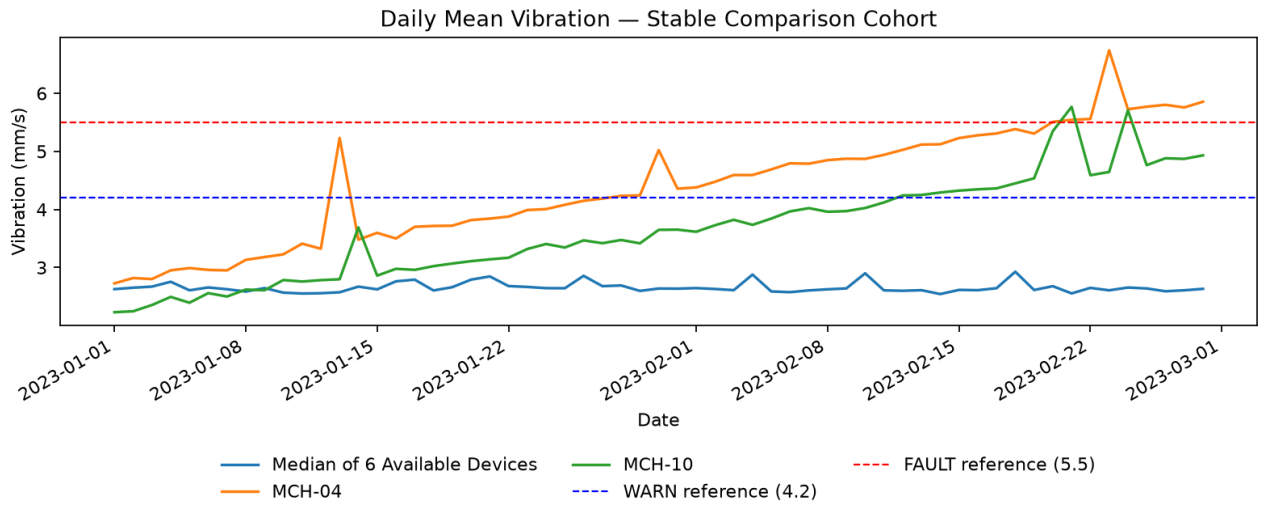


Figure 3: Compares daily patterns for `MCH-04` and `MCH-10` with a reference group of six devices having full daily coverage, including the reference mean for descriptive comparison.

7.4 Temperature–Vibration Relationship

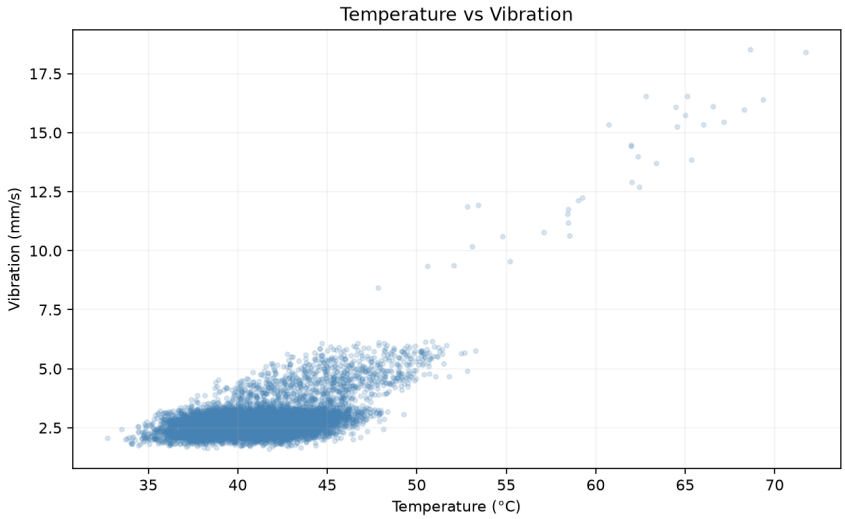


Figure 4: Displays an overall positive association between the two variables, consistent with the calculated Pearson correlation. Neither the scatterplot nor the correlation coefficient alone establishes causation.

8. Discussion and Key Findings

The findings show that assessing data quality involves more than removing duplicates and standardizing column formats. Recording quality, the choice of reference population, and statistical threshold definitions can all affect derived indicators and interpretations.

The principal findings are as follows:

1. **Validated cleaning actions:** Exact duplicates and formatting inconsistencies were corrected without unjustified imputation or deletion of outlier measurements.
2. **Concentration of elevated vibration:** `MCH-04` and `MCH-10` account for a substantial proportion of vibration outliers and strongly affect the mean for `NOR`.
3. **Sensitivity of outlier definitions:** IQR results change with the reference population used to derive limits; even equal totals can represent different flagged records.
4. **Subset-dependent correlation:** The full-dataset correlation is greater than the correlation calculated for `OK` readings alone.
5. **Gaps are not necessarily introduced by cleaning:** Long gaps already exist in the original synthetic data, while the estimate of potentially unrecorded readings assumes a regular sampling interval.
6. **Programmatic checks are not engineering verification:** Consistent file content, data types, and categories validate particular processing steps but do not independently prove real equipment failures.

9. Study Limitations

- The dataset is synthetic, not a record of actual factory operations; findings cannot be generalized to physical equipment without external validation.

- Independent device and sensor specifications are unavailable to establish safe operating limits or explain high readings.
- The `FAULT` label is not proof of a documented mechanical failure.
- IQR outliers depend on the reference population and chosen limits; an outlier is not necessarily an error.
- Estimated unrecorded readings rely on observed time intervals rather than a verified official sampling schedule.
- Correlations and daily trends do not establish causation or predictive capability.
- The project lacks independently verified failure-event records and their timestamps; no machine-learning model was trained.

10. Recommendations and Future Development

The findings support the following recommendations when working with comparable industrial data:

1. **Preserve reference and raw data:** Maintain traceability and the ability to repeat cleaning and comparisons.
2. **Document cleaning decisions:** Distinguish formatting problems from statistically unusual but potentially valid measurements.
3. **Use approved operating limits where available:** Do not treat statistical outliers as actual failures without engineering context.
4. **Verify the sampling policy:** Distinguish operational recording gaps from acceptable intervals between readings.
5. **Review the origin of status labels:** Establish their meaning before using `status` as a supervised classification target.
6. **Check reproducibility:** Rerun both notebooks and compare generated results and saved files with the documented findings.

Future potential: The cleaned data may be used for later **machine-learning experiments**, such as anomaly-pattern analysis. Developing a reliable predictive-maintenance model would require a well-defined target, suitable data about failure events and their timing, and an evaluation design that prevents future-information leakage (**data leakage**).

11. Project Files and References

Component	File
Data-quality assessment and analysis notebook	`01_data_quality_analysis.ipynb`
Data-cleaning and validation notebook	`02_data_cleaning_validation.ipynb`
Original data	`original_data.csv`
Working raw data	`raw_data.csv`
Cleaned data	`cleaned_data.csv`
Charts	`charts/`
Project description and reproduction instructions	`README.md`
Execution environment	`requirements.txt`

References:

- Datanemics — Factory Sensor Telemetry — Dataset source and technical description.
- Creative Commons — CC0 1.0 — License text stated by the source.
- Accompanying project notebooks and CSV files — Source of calculations, analytical results, and validation procedures.

12. Conclusion

The project completes an integrated cycle of **data understanding, quality assessment, cleaning, validation, analysis, and visualization**. Validation tests show that the cleaned data and analytical data are equivalent under the specified comparison conditions and that the primary numerical findings can be reproduced from the project files.

Differences among devices, temporal gaps, and statistical outliers merit further investigation, but they do not independently demonstrate operating causes or diagnose failures. The project thus provides **a documented foundation for industrial data analysis** and a starting point for more advanced experiments once suitable data and engineering criteria become available.